

1 Computer Network and The Internet = 1966 classmate

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Computer Network

→ A group of computer / electronic device which are connected to each other for the purpose of sharing their resources is called computer network.

World First Network → ARPANET (1969)

→ ARPANET: Advance Research project Agency Network

Characteristics of computer Network:

i) Resource sharing

- Allows sharing of hardware (printer, servers), software and data among across networks.

ii) Communication speed

- Facilitates data exchange, messaging, video calls, and collaborative work between users

iii) Backup

- All the data is retrieved if the device or connection is corrupted

iv) Scalability

- Networks can expand by adding more devices, users, and components without major changes

v) Reliability

- Provides fault tolerance and backup paths to ensure continuous operation even during failures.

V) Security

- Implements access control, encryption, and authentication to protect data and network resources.
- Cost effectiveness
- Performance
- Flexibility

Types of Computer Network:

I) PAN: Personal Area Network

- PAN is a short-range network designed to connect devices around a single person.

Ex: Bluetooth, Hotspot, etc.

Range is about ≤ 10 meters.

use case: In office, home, etc.

Characteristics:

- keeps your stuff synced and portable
- Don't support for long range, get's disconnected.

II) LAN: Local Area Network:

- It is a communication network that interconnects a variety of devices and provides a means for information interchange among those devices.

Ex: wifi, server,

Range is about ≤ 150 meters

use case: school, college

Characteristics

- Each connected device has the potential to interact with another device on the network.
- Some security should be executed if it is essential to protect private data.

III) MAN : Metropolitan Area Network:

- It is designed to extend over an entire city. It may be single network capable to cover a district.

MAN = LAN + WAN

Eg: SIM card, etc

- Range is about ≤ 50 km
- use case : District, state, private company

Characteristics:

- conversion from LAN to MAN is easy
- technical people required to set up.

IV) WAN : Wide Area Network.

- A network that spans several sites across a city, country or even the world is called WAN.

Eg: Geographical area, Aviation

Range is unlimited

WAN = MAN + LAN + PAN

Characteristics

- Multiple users can connect together in the world.
- server down and disconnection issues.

Advantages of Computer Network:

- i) Open to Everyone
- ii) Easy to add new devices
- iii) Resources sharing
- iv) Saving money
- v) Backup & storage
- vi) High Reliability
- vii) Scalability
- viii) Security

Disadvantages of Computer Network:

- i) Network device required
- ii) Virus attack
- iii) Required Handler
- iv) High speed internet
- v) Server controls
- vi) Management team

Mode of Communication Flow:-

1) Simplex: In simplex mode, communication is unidirectional.

→ In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive.

Keyboards and traditional monitors are both ex of simplex devices. Also Radio News.

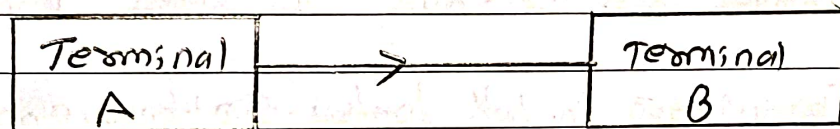


Fig: Transmission in one direction.

2) Half-Duplex:

→ In half-duplex mode, each station can both transmit and receive, but not at the same time.

- When one device is sending, the other can only receive, and vice versa.

Eg: walkie-talkie

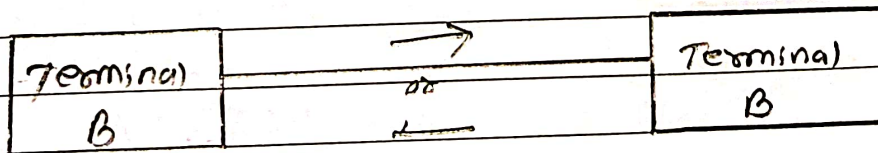


Fig: Transmission in either direction, but not simultaneously.

g) Full-duplex:

→ In Full-duplex mode, both stations can transmit and receive simultaneously. This model is like a two-way traffic flowing in both directions at the same time.

Example: Telephone network, online chatting, social media apps

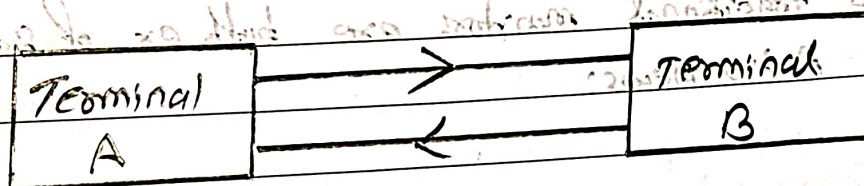


Fig: Transmission in both direction simultaneously.

Network connecting Devices:

There are several network devices that are the part of a computer network, each device performs a different function.

Some devices are

- i) Hub
- ii) Switch
- iii) Router
- iv) Bridge
- v) Gateway
- vi) Repeater
- vii) Modem
- viii) NIC
- ix) IR

① Hub:

→ It is a network device that is used to connect multiple computers in a network. All the information sent to the hub is automatically sent to each port of device (No private use for broadcasting).



- Hub is less expensive, less intelligent and less complicated.
- Transmission mode of Hub is "half duplex".

Advantages:

- The hub can broadcast the message.
- It is less expensive that anyone can use it.
- Easy installation.
- Robust: Limited scenarios, inefficiencies.

Disadvantages:

- If the hub fails, entire network fails.
- we cannot send private message.
- Hub cannot provide any security.
- Hub cannot support full-duplex transmission mode.

② Switch:

- It is a network device that connects multiple computers. It is mainly used to send the private data as well as there is not working of data.
- Switch is an advanced form of the hub similar in function but with changes.
- Switch can easily identify that which device connected with which port by using MAC Address.
- It delivers message on particular destination machine.

Advantages:-

- It is generally used to unicast message.
- It provides more security than hubs.
- It supports full duplex data transmission.
- If a node fails there will be no effect in the entire network.

Disadvantages:-

- If switch fails entire network fails.
- It is more expensive.
- Difficult to setup.

③ Routers:

- Compared to hub and switch router are smarter. Routers use a more complete packet "address" to determine which router or work station receive each packet next.

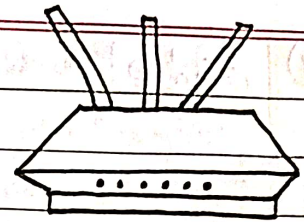


Fig: router

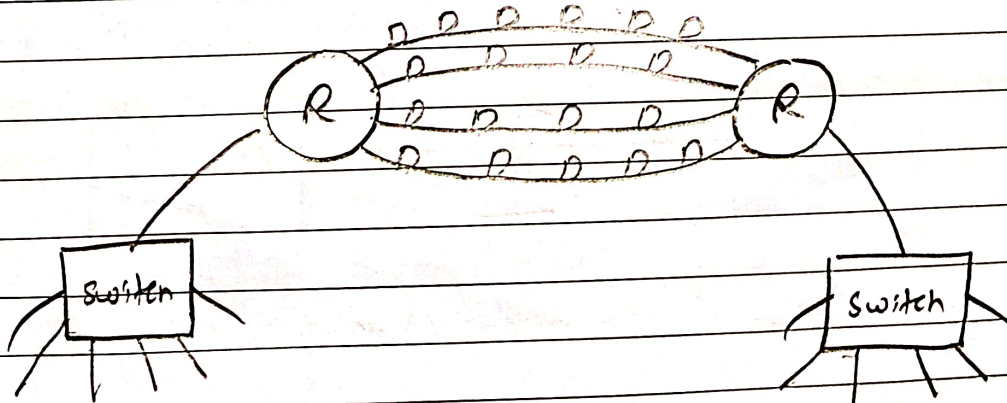
- The main work of router is to choose the congestion free path through which the data packet will travel from sender to the receiver.
- Router receive data packet to the sender, and for Analyse and Forward to the receiver.

Advantages:

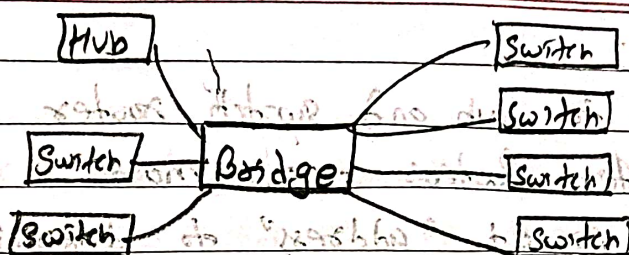
- It provides connection between two similar kind of network.
- Transmission rate is very high.
- It provide both wired and wireless connection.

Disadvantages:

- Router is more expensive as compared to other network device.
- Routers are complex to maintain.
- It only work with routable protocol.



IV) Bridge:-



- A bridge in computer networking is a device that connects and filters traffic between two or more network segments, making them function as a single network at the data link layer.
- Advantages:-

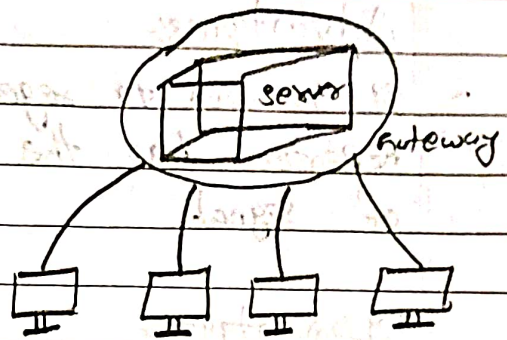
- Reduces network traffic by filtering and forwarding data only when necessary.
- Increases overall network performance and efficiency.

Disadvantages:-

- can become a bottleneck if overloaded with traffic.
- Doesn't work well with large or complex networks compared to more advanced devices like switches.

⑦ Gateway:

- Gateway is a network device i.e. used to connect two different kind of network. It allows to send and receive data through the internet even if it is a LAN network.



- It operates all seven layers of OSI model.

Advantages

- It connects two networks which have different protocols.
- We cannot access the internet without a gateway.

Disadvantage:

- It is more expensive.
- Difficult to maintain as well as very complex.

⑥ NIC

It is a network device without which we cannot connect computers to the internet/network.

⑦ Repeater:

- Repeater is a network device through which we can boost up the weak signals.
- When the signal travels in the network after travelling some distance the intensity of signal becomes low.
- In order to regenerate the weak signal we should use repeater device. It provides wired and wireless facility.

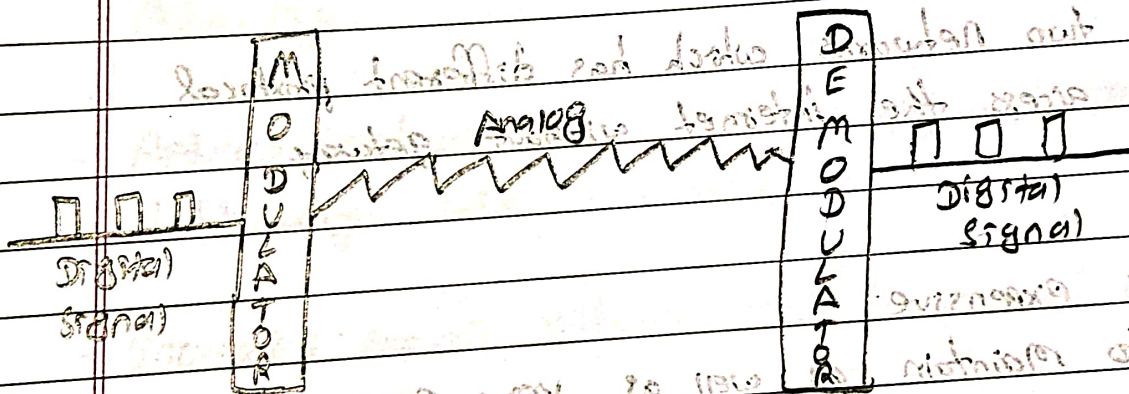
Advantages:

- It is used to regenerate the weak signal.
- Repeater has the ability to extend the length of signal.

Disadvantages:

- It require number of repeaters after some distance.
- They cannot receive network traffic.

(VIII) Modem:



Network Topologies

→ It refers to how devices in a computer network are connected. Network Topology is the way that defines the structure and how these components are connected.

Characteristics

(1) Physical topology

→

No. of port
No. of cable
Protocol

CSMA = carrier sense multiple access

CA = collision

CD = collision detection

classmate

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Types of Topology

- ① Point to point topology
No. of Port = No. of node
No. of cable = N
- ② Ring Topology : No protocol, use MAC address & R-switch
- ③ Bus Topology : Backbone cable / Protocol : CSMA/CA / CSMA/CD
No. of Port = No. of node
No. of cable = N+1
- ④ Star Topology : Protocol : Token Ring
- ⑤ Mesh Topology : No. of Port = No. of Station $\times$ 2
No. of cable = $\frac{N(N-1)}{2}$
- ⑥ Tree Topology : Protocol : DHCP (Dynamic host configuring protocol)
- ⑦ Hybrid Topology : combination of more than one topology

OSI

- Open system interconnection Developed by $\rightarrow$ ISO (International organization for Standardization) in 1984
- Application Layer
- Presentation "
- Session "
- Transport "
- Network " $\rightarrow$ logical link control sub layer
- Data link " $\rightarrow$ media access control sub layer
- Physical "

Working of OSI layer

- 1) Physical Layer
wire / wireless / transmission media / topologies
layer components
- 2) Data link Layer:
 $\rightarrow$ Node to Node delivery (source mac + destination mac) + error correction

III Network Layer

- Same IP + Destination IP + Packet delivery

IV Transport Layer

- Segmentation + TCP (No loss) + UDP (may loss) + Port Address + Point to Point communication + Error Recovery

V Session Layer

- Session (create + manage + destroy) + message send successfully

VI Presentation Layer

- Translation, + Encryption, + description + Authentication.

VII Application Layer

- user to user interaction + service provided by user + All application and software worked in this layer.

Network devices used in OSI layer.

Application layer

Presentation layer

Session "

Transportation "

Network "

Data Link "

Physical layer

Gateway

Router

switch / bridge

Hub / repeater

Data unit transfer

Software layer	Application Layer	:] Data	Application specific address
	Presentation "		
	Sessional "		
Header	Transport "	:] Segment	→ port address
	Network "		→ IP address
Hardware layer	Datalink "		→ MAC address
	Physical layer		→ No address

Address used in OSI layers: